Insect Detection and Classification on Soybean Crop

Praveen Thakur
191000036 (CSE)
IIIT Naya Raipur
Praveen19100@iiitnr.edu.in

Satyapal Singh Kanwar
191000045 (CSE)
IIIT Naya Raipur
satyapal19100@iiitnr.edu.in

Dhananjay Manjare
191000014 (CSE)
IIIT Naya Raipur
dhananjay19100@iiitnr.ed

Dr. Anurag Singh
Assistant Professor, ECE
IIIT Naya Raipur
anurag@iiitnr.edu.in

***Abstract*— The usage of cutting-edge technical solutions in Indian agriculture to address different farming-related concerns, such as crop health, weed difficulties, crop illnesses, etc., is currently behind. By offering technical solutions to automatically identify insects in soybean fields, we want to close this gap. The soybean (Glycine max) is an annual legume in the pea family that produces edible seeds (Fabaceae). The soybean is the most commercially significant bean in the world, making components for hundreds of chemical products in addition to serving as a source of vegetable protein for millions of people. A computer vision job called object detection includes identifying an object type and its position in an image. We used a well-known object detection method to identify insects in soybean crop areas. The YOLO v5 system has been trained to recognize and identify the presence of insects on the field. According to the simulation findings, YOLO v5's insect detection accuracy was 97% mean average precision (mAP).**

Keywords*— *Object Detection, Soybean Insect Identification, YOLO v5, Deep Learning, Internet of Things

I. INTRODUCTION

The majority of people in India make their living from farming, but because it is mostly carried out using antiquated agricultural techniques and instruments, average productivity is only moderately high. Agriculture in India is a non-attractive job since there is a severe lack of innovation and technical equipment utilization. It is also to blame for the farmers' low incomes and low crop yields. Low harvests caused by recent recurring disasters like drought, flood, insect infestation, etc., have actually hampered the development of farming. The main method for dealing with pest management is the use of synthetic pesticides [11]. It is challenging to coordinate successfully with the associated pesticide kinds since there is a lack of specialist information on large rural regions. Pesticides are thus applied more often and ineffectively. Because pest presence may be determined with a constant evaluation of their infestation pattern, the Integrated Pest Management (IPM) [1] theory claims that early and continual observation of the insets can enhance the utilization of pesticides in particular infected regions. This can lessen the adverse effects on the economy, the ecology, and the natural world. However, as this requires manual investigation, it has a number of drawbacks, such as being time-consuming and labour-intensive. Additionally, manual identification and evaluation are unreliable. In order to replace traditional farming practices [8], it is crucial to develop an accurate and constant crop monitoring approach. There haven't been many recent efforts on crop health monitoring, forecasting, and yield prediction described in the literature up to this point [10]. The research showed a machine vision application that used the loss method for initial pest detection and then scale-invariant feature transform (SIFT). Since it is possible to discriminate between six different kinds of creepy crawlies, it takes into consideration more accurate pest identification. The scientists used an SVM arranging approach to identify thrip insects in photos of yield shelters [3].

II. LITERATURE REVIEW

Early insect identification and detection can significantly increase crop output and quality [6] while also aiding in pest management. In the area of computer vision, various computer-based methods for identifying pests have recently been introduced. It may also be categorized as handmade feature extraction and deep learning feature extraction in [7]. Handcrafted features are a fantastic way to express low-level features. Early approaches were mostly dependent on handmade elements. Six hundred nine photos from eight classes of tea insects are identified by the author in [6] using correlation-based feature selection and ANN [4]. In order to recognize thrips, whiteflies, and aphids on leaves, an SVM classifier [4] was utilized. These approaches often relied on a small number of typical hand-tailored traits to identify insects using extremely few datasets and a variety of insect classifications. There are many bug classes, though. Using extractors to see grouped insect classes is cumbersome and inefficient. Additionally, hand-tailored characteristics for semantic data have a constrained capacity for depiction. Learning-based strategies have drawn the interest of researchers in this sector recently due to their superior performance. Deep convolution neural networks, Google Net, and ResNet, as examples, display excellent performance in the picture categorization problem. A few works have also successfully used CNNs to address the problem of recognizing creepy crawly bugs. In order to categorize insects found in paddy fields, a deep CNN is trained on 5000 training samples divided into twelve groups. The categorization of Safe Net for paddy pests is explored. Additionally, the authors get successful results using a bio-inspired approach on the deep CNN known as VGG Netto [9]. Despite this, there are only 563 samples in the evaluated dataset. In these deep feature-based efforts, there aren't enough samples to reliably maximize the enormous hyper parameters of CNNs.

III.DL MODEL FOR OBJECT DETECT

A. YOLO v5

In the fields of particle physics, data science, and artificial intelligence, Ultralytics has been leading various U.S. Intelligence Community (IC) and Department of Defence (DoD) efforts since its establishment in 2014 [4]. On a number of productive projects, Ultralytics has collaborated with a number of universities, national labs, and other start-ups. The most sophisticated Vision AI in the world, YOLO v5, was just developed by Ultralytics. With YOLO v5 and its Py Torch implementation, object detection is simple to begin using.

B. CNN

A Deep Learning algorithm is a convolutional neural network, sometimes referred to as a CNN or ConvNet. It begins by classifying the complete input picture after giving the image's individual components weights and biases. ConvNets may learn filtering and classification with enough practice, and they require less pre-processing than other techniques [5]. The unique architecture of artificial neural networks is the convolutional neural network (CNN). Some visual cortex properties are used by CNN. Image categorization is one of the most often used applications for this architecture [5].

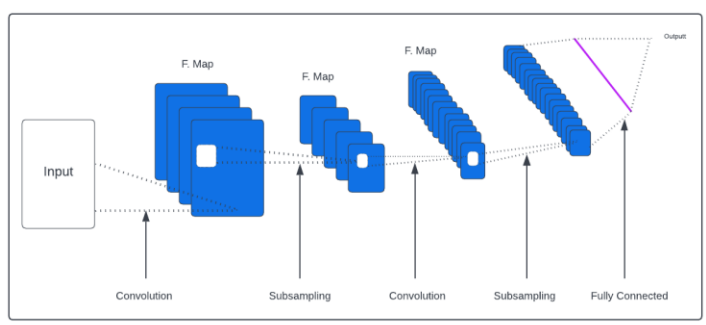

Fig 1: Convolutional Neural Network

C. SSD

The SSD is based on the use of convolutional networks that generate numerous bounding boxes of different fixed sizes and score the presence of the object class instance in those boxes. The final detections are then produced by a non-maximum suppression step after the convolutional networks have finished their work [12].

According to how the SSD model operates, each input picture is split into grids of varying sizes, and at each grid, detection is carried out for various classes and various aspect ratios. Each of these grids is given a score that indicates how well an object fits in that specific grid. The final detection is then extracted from the set of overlapping detections using non-maximum suppression. The SSD model's fundamental premise is as follows.

It has been demonstrated that the SSD model outperforms earlier state-of-the-art detection methods like YOLO and Faster R-CNN [12].

The SSD model is one of the fastest and most efficient object detection models for multiple categories.

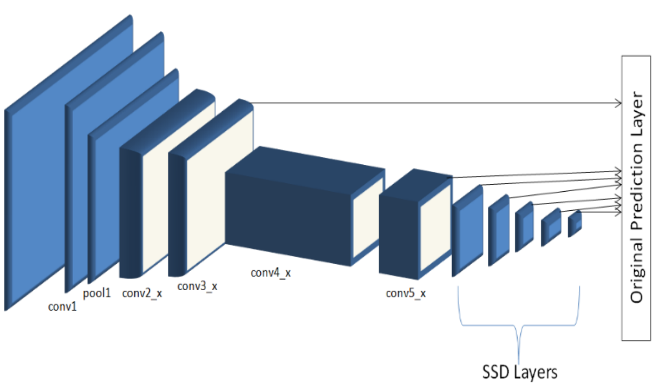

Fig 2: SSD

IV. METHODOLOGY

A. System Background

Various illnesses and insect infestations can affect soybean plants. Numerous variables, including biotic diseases brought on by environmental elements like temperature, humidity, excess nutrient (plant food), light, plant species, and pests that spread the disease from one plant to another, can have an influence on crops.
Considering the information above, we concentrate on two factors in our study, namely [4].

1. Insect identification in the crop.
2. Identifying the location of the insect in the image

B. System Overview

The suggested work's object identification algorithm intends to identify five different insect species on the soybean crop that may have a big impact on yield—the system uses a high-level block diagram for detection. The suggested system uses raw data, followed by data pre-processing and image annotation or labelling. The data set is then split into training and test sets. The train set is used to train YOLO algorithms once the data has been divided into a test and train set. The system can now identify five types of soybean insects that were provided throughout the training phase. We are currently evaluating our system using the test set.

C. Dataset

We used data from Raipur, India's Indira Gandhi Krishi Vishwavidyalaya.

There are five distinct insects in it;

1. Aloa lectinea
2. Eocathecona furcellata
3. Pod borer
4. Larva spodoptera
5. Rice ear bug

D. Data Flow Diagram

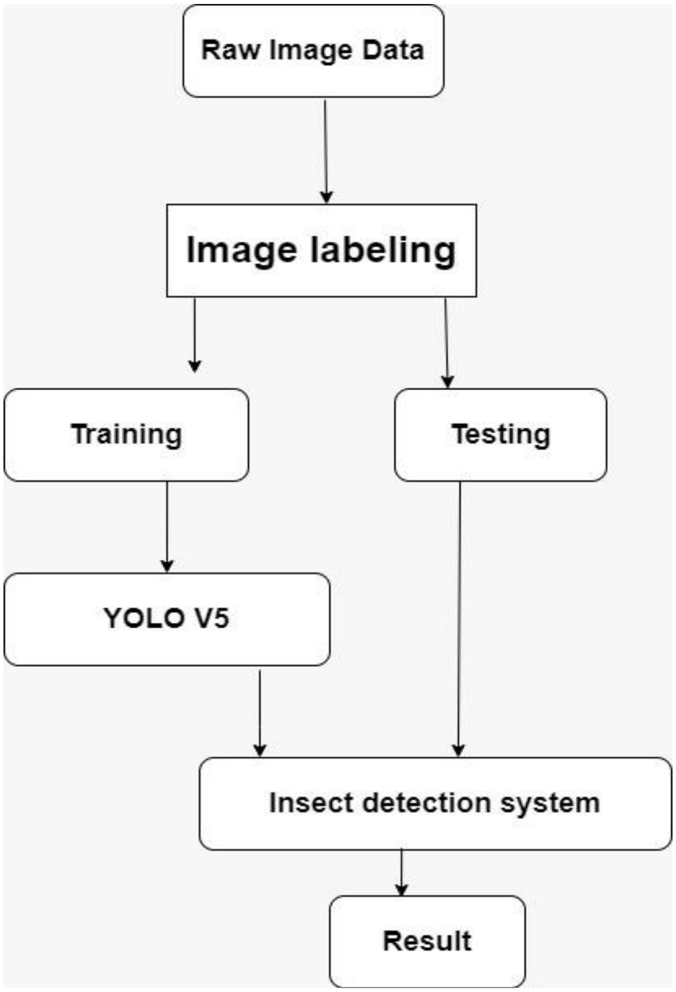

Fig 3: Data Flow

E. Android Implementation

For the implementation of SSD model in Android, we used the interface provided by the "TensorFlow Android camera demo”. First, we trained the SSD model Using Transfer Learning. Then we converted the trained SSD model to the TensorFlow model using tf-nightly (TFLite converter works better with tf-nightly). Then we converted TensorFlow saved model to the TFLite model using python API. Finally, we created TFLite metadata for the TFLite model which is the required Tiny model that can be deployed on Android phones to detect insects. After that, we Loaded this TFLite model on the Android app. The app is designed with a streaming video from the camera, and each image frame is passed to the SSD model for insect detection. And then the detected results are marked with boxes with their name in real-time.

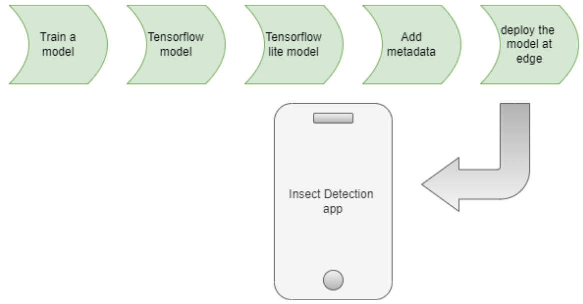

Fig4 : Application Flow Diagram

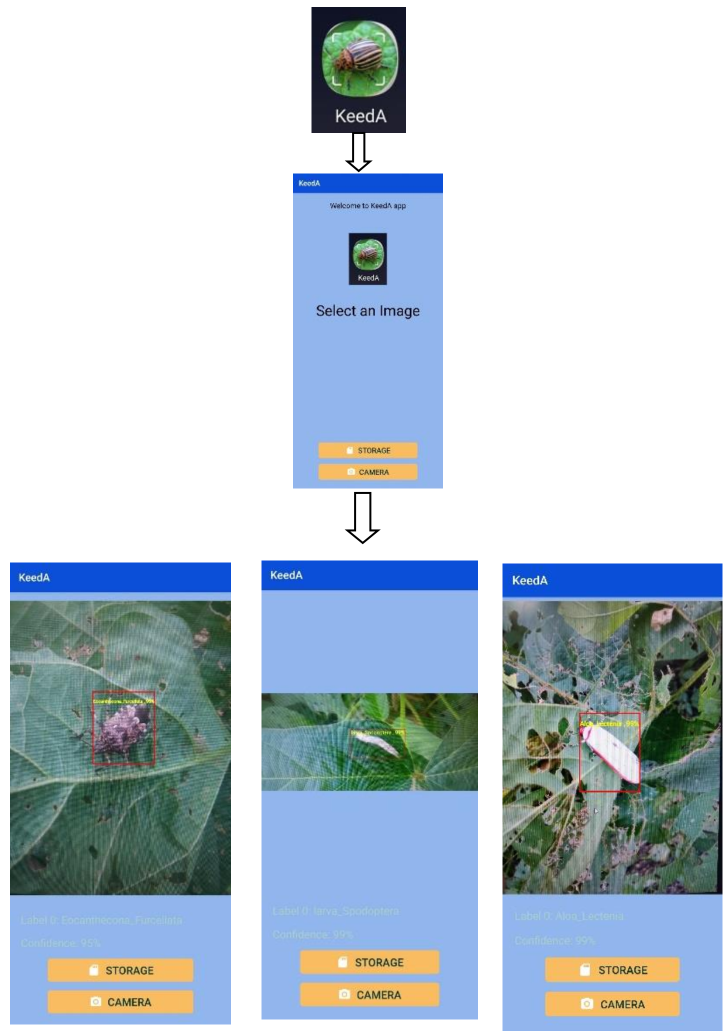

Fig5 : Application Interface

V. RESULT

The findings and observations of the tests done on various models are detailed in this section. Following the testing of the trained models, it can be concluded that models trained on segmented pictures outperform colour and greyscale images. The images have very less noise, which also contributes to the high accuracy of the models.

We have adopted the following settings for the experiment:

1. Input Size: The input size of images for the models trained is 416x416.

2. Split: A training set and a testing set are created from the dataset in an 80:20 split. The splitting is done in a random fashion. The training set consists of 2700 images, while the test set contains 450 images.

3. We have used the Stochastic Gradient Descent (SGD) optimizer for training the YoloV5 model.

A. Using YoloV5

Modern object detection algorithm YoloV5 is quick without losing too much accuracy.
The proposed framework is trained using PyTorch deep-learning library and is measured using performance metrics which are as follows:

1. mAP

mAP is the accuracy when IOU is greater than or equal to the threshold.IOU or Intersection over Union is the ratio of the area of intersection of a true and predicted region of interests to the union of the same. Following are the different values of mAP at different thresholds.

A. mAP@0.5: The average precision of the proposed model at 50% confidence is 98.776%.

B. mAP@0.5:0.95: The mean average precision of the proposed model at 50% to 95% confidence is 71.594%

2. Precision

The capacity to only recognize important items is precision. It is defined as the proportion of positively classified outputs that are accurately categorized as all of the positive outputs. Mathematically,
Precision = TP/(TP + FP)
Where
A. TP is True Positive.
B. FP is False Positive.

3. Recall

The ability of a model to properly identify True Positives is measured by its recall. It is defined as the proportion of properly categorized outputs to correctly classified positive outputs.

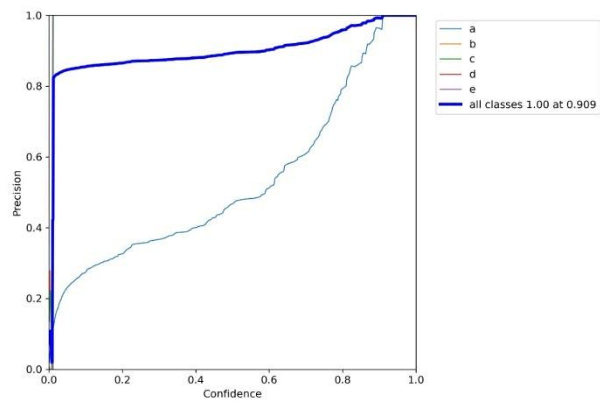

Fig 6: In this graph Precision increase with confidence drastically at the start and then increases gradually

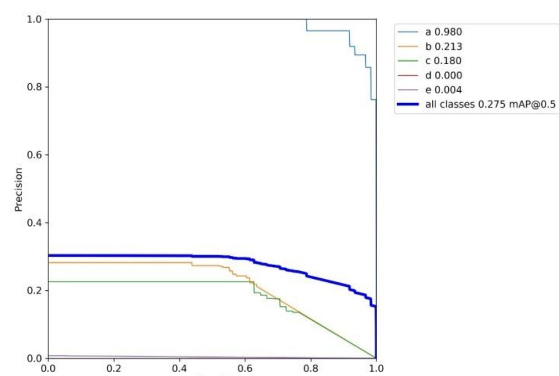

Fig 7: In this graph, recall start to decrease precision after some point on different classes for the insect datasets used

B. Using SSD

1. Loss: The reason for this loss happens because of the loss function, which includes both classification and localization loss.

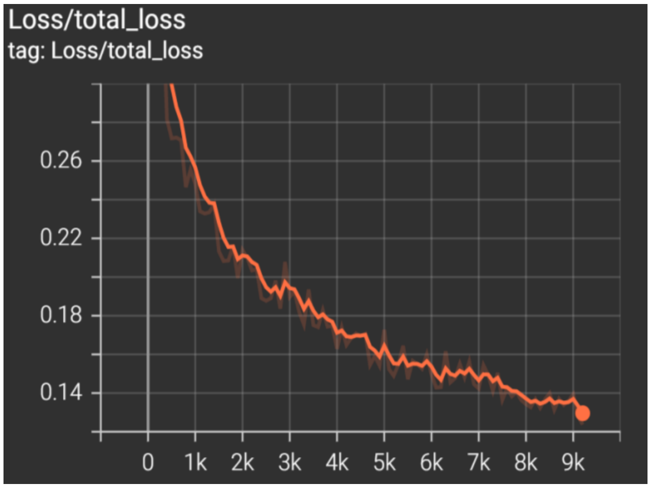

Fig 8: Total Loss during training

2. Regularization loss: It is just the additional loss caused by the regularisation function, which modifies the loss function by penalizing certain weight values as you learn them.

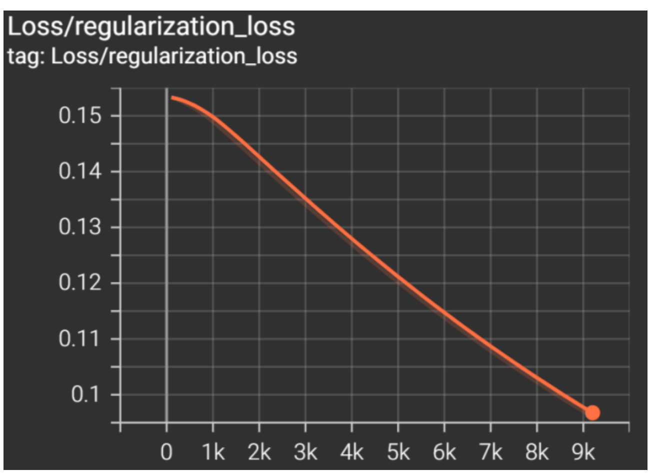

Fig 9: Regularization loss during training

3. Classification loss: In order to train the classification head to identify the kind of target item, this was used.

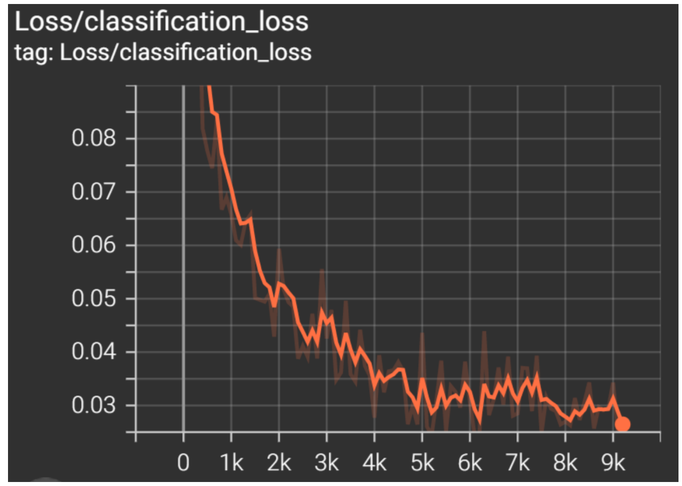

Fig 10: Classification loss during training

4. Localization loss: This is used to teach a different head to locate the target item by regressing a rectangular box.

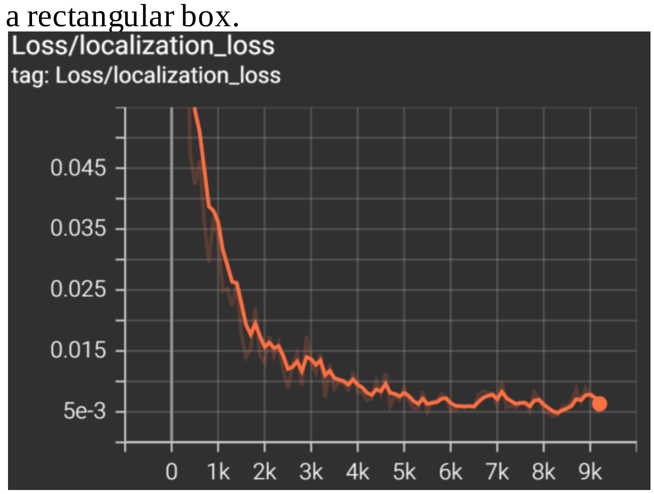

Fig 11: Localization loss during training

D. Detection of Insects in Real Time with TensorFlow:

Deep learning is a subset of machine learning that uses sensory networks that stimulate the structure and function of the human brain. In this paper, we use TensorFlow, an open-source library that Google develops.

With the help of this library. We can deploy the app. To implement the TensorFlow application for general use first step is to create the TensorFlow lite model, and after implementing this, we have to generate the meta-data for our model, whose extension is. TFLite; after this, we have to provide the class name in txt format after step just run the app.

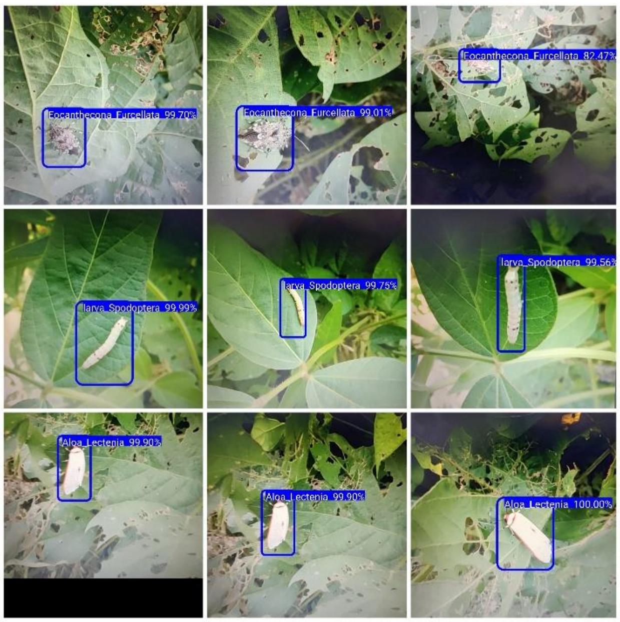

Fig 12: Application Result

VI. CONCLUSION

In this study, three deep convolutional neural network techniques are investigated for insect detection and identification using a soybean insect dataset that we produced. Three variants of these algorithms—Yolo v5, v4 tiny, CNN, and SSD—were used to detect insects since Yolo algorithms and SSD are well renowned for their precise detection skills in small objects. The effectiveness of Yolo and SSD as algorithms for finding and locating insects is shown through comprehensive simulation results. The SSD plus Yolo method appears to attain the highest mean average accuracy per class, according to quantitative data. It can recognize tiny things and is fairly quick at both training and detection. We can observe that Yolo v4 is small, and Yolo v5 are doing well when we look at the F1 score. Thus, we may draw the conclusion that SSD and Yolo v5 are more effective than Yolo v4 small and CNN in identifying agricultural insects and can be used to automatically identify insects in a variety of crops.

VII. REFERENCES

[1] E. N. Bernardo, "Adoption of the integrated pest management (IPM) approach in crop protection: a researcher's view.," Philipp. Entomol., vol. 9, no. 2, pp. 175–185, 1993.

[2] S. Khan, S. Nadir, G. Lihua, J. Xu, K. A. Holmes, and Q. Dewen, "Identification and characterization of an insect toxin protein, Bb70p, from the entomopathogenic fungus, Beauveria bassiana, using Galleria mellonella as a model system," J. Invertebr. Pathol., vol. 133, pp. 87–94, 2016, doi: https://doi.org/10.1016/j.jip.2015.11.010.

[3] C. Wen, D. E. Guyer, and W. Li, "Local feature-based identification and classification for orchard insects," Biosyst. Eng., vol. 104, no. 3, pp. 299–307, 2009, doi: https://doi.org/10.1016/j.biosystemseng.2009.07.002.

[4] S. Verma, S. Tripathi, A. Singh, M. Ojha and R. R. Saxena, "Insect Detection and Identification using YOLO Algorithms on Soybean Crop," TENCON 2021 - 2021 IEEE Region 10 Conference (TENCON), 2021, pp. 272-277, doi: 10.1109/TENCON54134.2021.9707354.

[5] Yamashita, R., Nishio, M., Do, R.K.G. *et al*. Convolutional neural networks: an overview and application in radiology. *Insights Imaging* **9**, 611–629 (2018). https://doi.org/10.1007/s13244-018-0639-9

[6] R. Samanta and I. Ghosh, "Tea Insect Pests Classification Based on Artificial Neural Networks," 2012

[7] L. O. Solis-Sánchez et al., "Scale invariant feature approach for insect monitoring," Comput. Electron. Agric., vol. 75, no. 1, pp. 92–99, 2011, doi: https://doi.org/10.1016/j.compag.2010.10.001.

[8] Steinke, J., van Etten, J. & Zelan, P.M. The accuracy of farmer-generated data in an agricultural citizen science methodology. *Agron. Sustain. Dev.* **37**, 32 (2017). https://doi.org/10.1007/s13593-017-0441-y

[9] K. Simonyan and A. Zisserman, "Very Deep Convolutional Networks for Large-Scale Image

Recognition," Sep. 2014, [Online]. Available: http://arxiv.org/abs/1409.1556

[10] Deep learning for crop yield prediction: a systematic literature review New Zealand Journal of Crop and Horticultural Science https://www.tandfonline.com/doi/ref/10.1080/01140671.2022.2032213?scroll=top&role=tab

[11] Jacquet, F., Jeuffroy, MH., Jouan, J. *et al.* Pesticide-free agriculture as a new paradigm for research. *Agron. Sustain. Dev.* **42**, 8 (2022). https://doi.org/10.1007/s13593-021-00742-8

[12] Ambika Neelopant , Dr. S. V. Viraktamath , Pratiksha Navalgi, 2021, Comparison of YOLOv3 and SSD Algorithms, INTERNATIONAL JOURNAL OF ENGINEERING RESEARCH & TECHNOLOGY (IJERT) Volume 10, Issue 02 (February 2021)